Fulcrum
Coffee Roasters



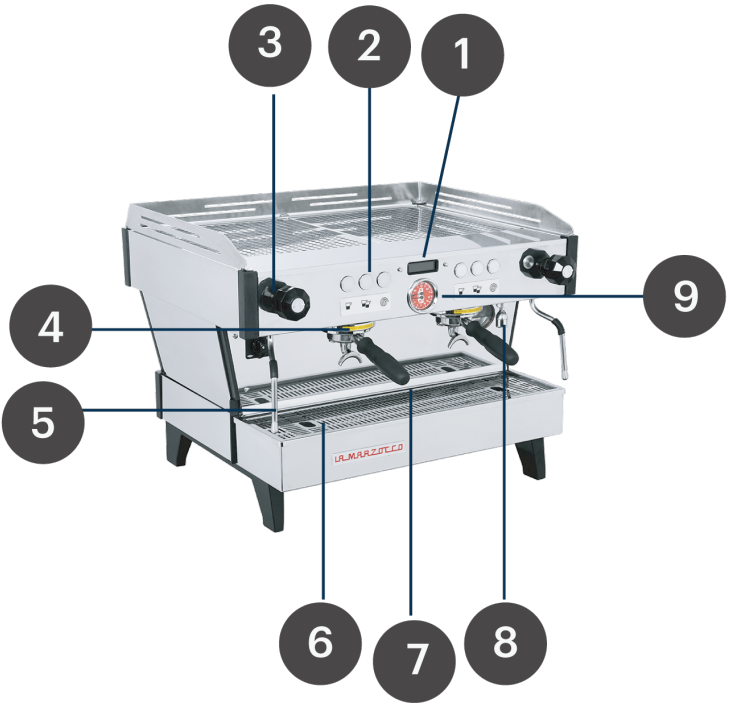
Barista Fundamentals



TABLE OF CONTENTS

01	MACHINE COMPONENTS	ESPRESSO MACHINE	PAGE 6
		GRINDER	PAGE 8
02	ESPRESSO BASICS	WHAT IS ESPRESSO?	PAGE 9
		GRIND	PAGE 10
		DOSE	PAGE 10
		DISTRIBUTION	PAGE 12
		TAMPING	PAGE 12
		BREW RECIPE	PAGE 14
		ESPRESSO PREPARATION STEPS	PAGE 15
		TASTE AND FINE TUNING	PAGE 16
03	MILK STEAMING		PAGE 18
04	ESPRESSO MACHINE DAILY MAINTENANCE		PAGE 20
05	GRINDER TYPES		PAGE 22
06	GLOSSARY OF DRINKS		PAGE 24

Espresso Machine



1 **LED DISPLAY**

Provides information about: shot timing, boiler temperature, volumetric pulses, etc. Provides menu adjustments and navigation through button presses.

2 **CONTROL SWITCHES**

Starts and stops brewing.

3 **STEAM KNOB**

Controls steam valve, allowing discharge through steam wand.

Espresso Machine

4 **GROUPHEAD**

Delivers hot pressurized water to the coffee in the portafilter.

5 **STEAM WAND**

Discharges steam from the tip.

6 **DRIP TRAY**

Catches waste water to be drained via the drain tube.

7 **PORTAFILTER**

Removable device housing the basket in which the coffee is held. Locks into place inside the grouphead.

8 **HOT WATER SPOUT**

Dispenses hot water from the steam boiler. Use caution, as it is typically too hot for consumption. Heavy use may drain the steam boiler, affecting steam wand pressure.

9 **PRESSURE GAUGE**

Displays current pressure of the main boiler and steam boiler, measured in bars. (Steam/boiler pressure: 1.2-1.8 bars, Pump pressure: 9 bars).

Grinder



1 **HOPPER**

Holds the coffee beans.

2 **ADJUSTMENT RING**

Adjusts the distance between burrs, altering the grind to be finer or coarser.

3 **START BUTTONS**

Engages grinder for specified dose times.

4 **POWER SWITCH**

Turns grinder on and off.

What is Espresso?

Espresso is a method of using pressure to force water through finely ground coffee. This produces a small amount of concentrated coffee and can be consumed on its own or combined to produce a wide variety of beverages.

In preparing espresso, there are two parties: The machine and the operator. The machine will perform the same operations identically every time. As the operator we must focus on how we use the machine and prepare the coffee.

Steps for Preparing Espresso:

1. Grind the coffee.
   a. Set the grind
   b. Set the dose.

2. Prepare the coffee.
   a. Distribute the coffee.
   b. Tamp the coffee.

3. Open the espresso machine
   a. Lock in the portafilter
   b. Extract espresso

Grind

The grind is a variable defining how fine or coarse the ground coffee is.

Adjusting the grinder finer moves the burr set closer together. An espresso that pulls slower due to a fine grind will be denser and more concentrated.

Adjusting the grinder coarser moves the burrs further apart from each other. An espresso that pulls quickly due to a coarse grind will be weak and watery.

Dose

The dose is a variable defining how much ground coffee (measures in grams) is being used to brew the espresso.

Select a dose based on the size of your portafilter's basket, generally +/-2g from your basket size.

Too much coffee inhibits water flow. Too little promotes channeling.

Figure 1

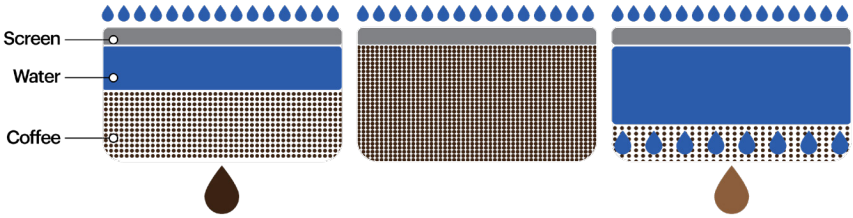


Figure 2



In figure 1, the grinder burrs are set further apart for a coarse grind.
In figure 2, the burrs are set closer together for a fine grind.

Figure 3



An ideal dosage of ground coffee.

Too large of a dose inhibits water flow through the bed of the coffee.

Too small of a dose will be prone to channeling.

Distribution

The ground coffee must be level and distributed evenly inside the basket. We can use a distribution tool in order to increase consistency, not only from shot to shot, but also from barista to barista.

Without a distribution tool, we can tap the sides of the portafilter to settle the grounds and allow them to fall evenly within the basket.

Tamping

The purpose of tamping is to compress the coffee and form a compact bed. This helps restrict the water flow at the correct pace and pressure, which allows for a more balanced extraction and flavor.

Tamp evenly, pressing down perpendicularly to the portafilter without tilting.

Tamp with roughly 20-30 pounds of pressure.
Make a 90° angle with the elbow of your tamping arm pointing up.

Figure 4

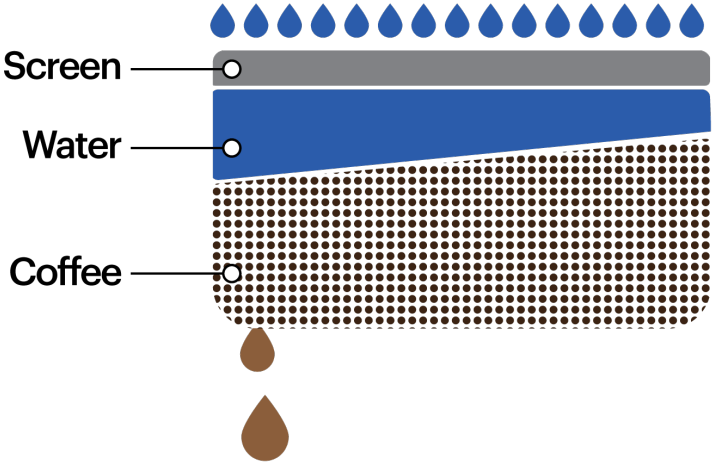
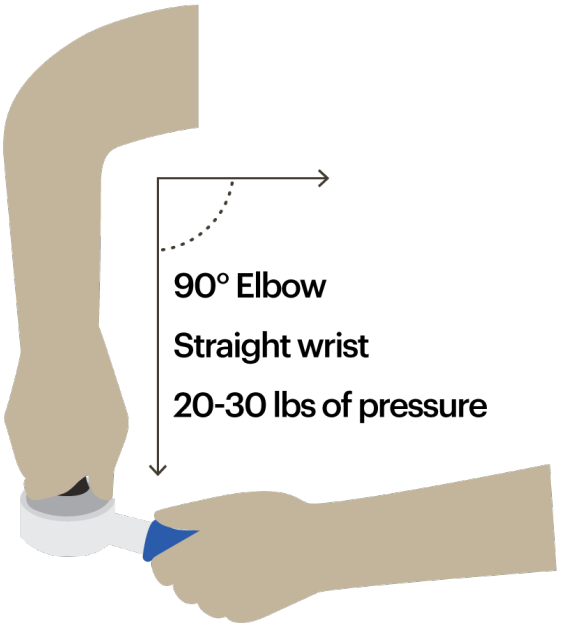


Figure 4 illustrates a bed of coffee distributed unevenly, resulting in an uneven extraction.

Figure 5



Brew Recipe

Much like cooking, preparing espresso needs a recipe, tools, fresh and good ingredients, and the knowledge to execute it.

So far we know how to dose and prepare our coffee. Next, we need to explore how much hot pressurized water to expose it to and for how long.

We recommend using somewhere between a 1:2 and 1:3 brew ratio of coffee:water, with a brew time between 20-35 seconds.

Using a gram scale, measure the brewed mass extracted to control how much water is used and a timer (usually in the espresso machine display) to measure how long the espresso extracts for.

EXAMPLE RECIPE:

Dose: 17 grams
Brewed Mass: 36 grams
Time: 28 seconds

IN A SHORTENED FORMAT:

17g / 36g / 28s

Coffee weight / Brewed Mass / Extraction Time

Preparation Steps

1. Remove the portafilter from the grouphead and rinse the basket by activating the grouphead briefly. This also clears coffee residue from the last shot out of the shower screen inside the grouphead.

2. Wipe the portafilter basket dry.

3. Grind coffee into the portafilter basket to the amount specified in your recipe. Use a gram scale to weigh the portafilter before and after grinding to ensure you're using a consistent amount of coffee with each shot.

4. Level the ground coffee using a distribution tool, or tapping the sides of the portafilter, or grooming the grounds using a clean finger.

5. Tamp the coffee using the proper technique (pg. 11).

6. Place your chosen vessel on a gram scale and position it underneath the portafilter spout. Lock the portafilter into place inside the grouphead and start pulling the shot by pressing the button activating the grouphead. Stop the shot once it reaches the brewed mass of the recipe.

7. Assess the espresso according to the recipe. If the shot pulled too quickly, adjust the grind finer. If the shot pulled too slowly, adjust the grind coarser. Taste your espresso and adjust grind based on the extraction principles on the next page (pg. 16-17).

8. Once you've attained an ideal espresso, identify and repeat the steps made to achieve it. The key to consistently good coffee is the consistent preparation of it.

Taste and Fine Tuning

Coffee is mostly water. The thing that makes coffee what it is, is dissolved coffee material suspended in water. As water passes through coffee it is dissolved partially (coffee is extracted).

How much of the coffee we extract or dissolve affects the flavor. Finer grinds restrict water flow allowing more extraction to occur.

How much water we use to reach our extraction affects the strength. This is why we recommend a 1:2 - 1:3 brew ratio for a desirable strength.

To fine tune our recipe to taste, we adjust the grind to control extraction. Taste your espresso to determine whether its flavor is indicative of under or over extraction.

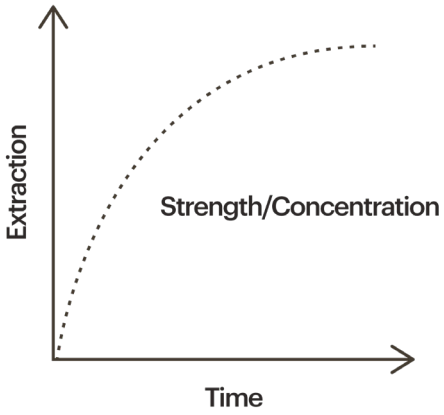
Grind finer to increase and coarser to decrease.

Figure 6



As extraction increases, the overall flavor of the espresso will change. Under extracted espresso will typically taste sour, over extracted espresso typically bitter, and well extracted espresso tending to be sweeter, or any of the intended flavors present in the coffee.

Figure 7



An espresso pulled using a coarse grind with 18g ground coffee to 36g extracted brewed-mass will pull very fast. Time, extraction, and strength for this shot will be low. The graph represents the relation of these variables as grind moves from coarser to finer.

Milk Steaming

Milk steaming is the incorporation of air into milk which creates micro foam. This is achieved through proper positioning of the steam wand, aeration, and temperature.

Aeration

Aeration at its basic definition is the mixing or dissolution of air into a substance. In our case we are incorporating air into milk.

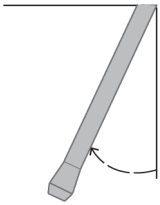
More air = more foam
Faster aeration = larger bubbles, stiffer foam
Slower aeration = smaller bubbles, dense, smooth foam

Steam Wand Maintenance

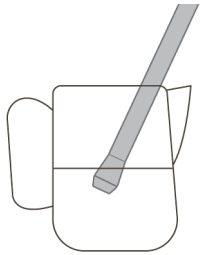
Make sure to purge the steam wand before and after steaming milk. This ensures the wand remains clean, since milk residue can remain stuck after use. Wipe it clean with a wet towel after each use.

Cappucino	Milk	Air	%Milk/Air
8 oz	4 oz	4 oz	50/50
12 oz	6 oz	6 oz	50/50
16 oz	8 oz	8 oz	50/50
Latte	**Milk**	**Air**	**%Milk/Air**
8 oz	6 oz	2 oz	75/25
12 oz	9 oz	3 oz	75/25
16 oz	12 oz	4 oz	75/25

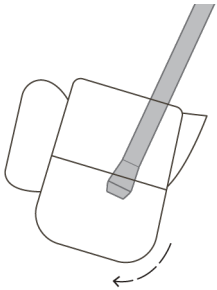
See glossary on pages 25-26 for diagrams.



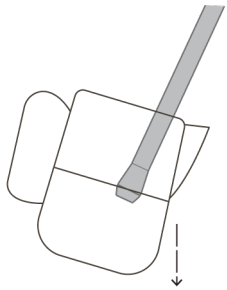
1. Position the steam wand at a 15° - 30° angle away from the espresso machine.



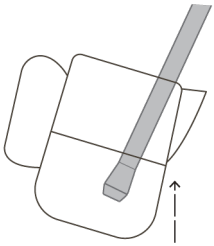
2. Submerge the tip of the steam wand in the center of the pitcher beneath the surface of the milk.



3. Tilt the pitcher slightly so the tip is submerged off center and causes a whirlpool when steaming.



4. Lower the steaming pitcher slightly to add air to the milk; stretching it and creating foam. The air entering the milk should produce a hissing sound.



5. Raise the steaming pitcher slightly to stop aeration.



6. Maintain a whirlpool, texturing the milk as it rises in temperature. Stop steaming once the milk has reached the desired temperature.

Espresso Machine Daily Maintenance

1. Take out the portafilters and separate the baskets.
2. Soak the baskets and portafilters in hot water and Puro caff.
3. Remove the screen and screw from the grouphead and clean with water and puro caff. Reinsert shower screen and screw.
4. Put about a teaspoon of puro caff into a blank portafilter basket. Add half oz of hot water and stir with espresso machine brush until melted. Lock portafilter into the grouphead.
5. Run the machine for 10 seconds, and rest 5 seconds. Repeat 5 times (back-flushing).
6. Rinse out the blank basket and lock it back in, repeating step 5 until the water from the grouphead runs clear.
7. Reinert the baskets into the portafilters.
8. Take out the drip tray and rinse clean. Leave machine ready for the next day.

Back



Flushing

Grinder Types

A wide variety of commercial espresso grinders may be encountered in a cafe. The grinder pictured on page 8 is one of the most common (Mazzer Electronic Grinder-Doser), but there are others available that have different features and capabilities. The following are a few that differ in purpose and capability.



Figure 8: Mazzer Major V

Electronic Doser

Grinds for a specified amount of time, with customizable presets.

Automatic Tamp

Tamps consistently and level every time.

Adjustable pressure from 20-60 lbs.



Figure 9: PUQ Press Gen5



Figure 10: La Marzocco Swift Grinder

Automatic Doser and Tamp

Grinds coffee to a preset volumetric dose. Distributes and tamps into portafilter automatically.

Glossary of Drinks



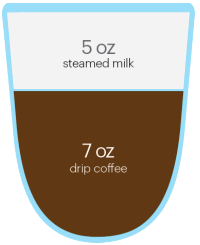
Espresso - *2 oz*
A small concentrated coffee beverage prepared by forcing hot pressurized water through finely ground, compact coffee.



Espresso Con Panna - *4 oz*
Espresso with whipped cream on top

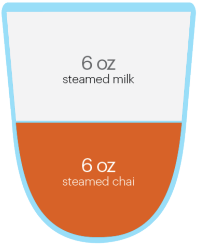


Americano - *12 oz*
A double shot of espresso poured over hot water. It can be served black or with cream & sugar, depending on the customer's preference.



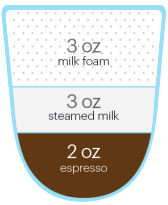
Café Au Lait - *12 oz*
Drip coffee with added steamed milk.

Milk Temperature:
145-155°F



Chai Latte - *12 oz*
Chai concentrate and milk of choice are combined in a milk pitcher then steamed like a latte. Refer to your chai concentrate's recipe for the correct ratio of concentrate:milk.

Milk Temperature:
145-155°F



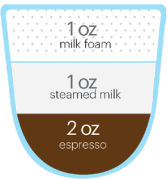
Cappuccino - *8 oz*
Double espresso with foamy, steamed milk. Milk is steamed to a lower temperature for optimal sweetness. Traditionally served in a smaller size (5 to 8 ounces). Larger cappuccino will be similar to a foamy latte.

Milk Temperature:
130-140°F



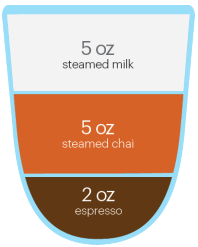
Cold Brew Coffee - *16 oz*
Batch coffee brewed overnight with cold water in a cold brew coffee maker. The cold extraction of the coffee tends to bring out a sweet, rich body that is low in bitterness and acidity. Typically drank black or with milk and sugar / syrup.

Recommended Product:
Fulcrum Cold Brew Concentrate



Cortado - *4 oz*
Equal parts espresso and steamed milk. Typically served in a 4 oz Gibraltar glass.

Milk Temperature:
130-140°F

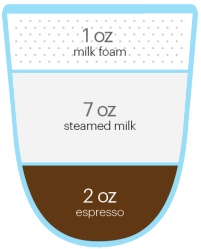


Dirty Chai Latte - *12 oz*
A chai latte poured over a double shot of espresso.

Milk Temperature:
145-155°F



Drip Coffee - *12 oz*
Batch brewed filtered coffee, usually brewed for either a half gallon or a gallon at a time. Drip coffee will require its own brewer and a separate grinder, as it is ground coarser than espresso.



Flat White - *12 oz*
An Australian-inspired small latte that uses a ristretto espresso shot. In the US, many cafes make the drink with a double espresso shot, and steamed milk that has lightly less foam than a latte, and a cooler temperature.

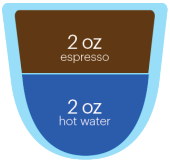
Milk Temperature:
130-140°F



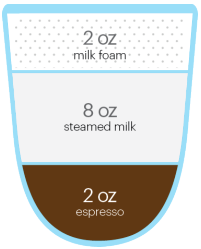
Hot Chocolate - *12 oz*
Steamed milk poured over & incorporated with chocolate sauce. A kid's hot chocolate will typically be in a smaller size, with the milk steamed to only a slightly warm temperature.

Milk Temperature:
110-125°F (Kids) | 145-155°F (Adult)

Recommended Product:
Torani Dark Chocolate Sauce



Italiano - *4 oz*
A smallller Americano. Equal parts espresso and hot water, typically 4 oz in size.



Latte - *12 oz*
Double espresso with steamed milk poured slowly on top. Milk will be aerated 25% and should have a silky texture. Flavors are commonly added, and a latte will be the base drink for most signature beverages.

Milk Temperature:
145-155°F



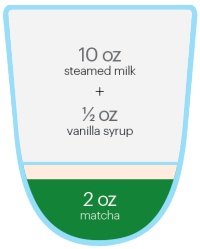
London Fog - *12 oz*

Earl grey tea latte originating from the pacific northwest, with vanilla syrup and steamed milk.

Tea Steep Time (before adding milk):
4 minutes

Milk Temperature:
130-140°F

Recommended Products:
Rishi Organic Earl Grey Tea
Republic of Tea Earl Greyer
Stash Earl Grey
Torani Vanilla Syrup
Torani Puremade Signature Vanilla



Matcha - *12 oz*

A latte made with matcha green tea powder as the base, instead of coffee. The matcha is whisked with 2 oz of water (185°f or lower) for 30 seconds or until incorporated. Customers will take a matcha latte with or without sweetener. Common sweeteners include vanilla and honey. Matcha lattes are also commonly used as the base for many matcha signature beverages.

Milk Temperature:
145-155°F

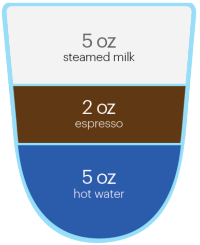
Recommended Products:
Rishi Organic Matcha Tea Powder
Organic Matcha DNA Powder
Torani Vanilla Syrup
Torani Puremade Signature Vanilla
Wildflower Honey Jug



Macchiato - *3 oz*

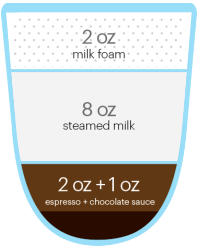
A double shot of espresso topped with 1 oz of foamy milk poured or scooped on top. Macchiato means “stained” or “spotted” in Italian.

Milk Temperature:
130-140°F



Mezzo - *12 oz*
Americano with added steamed milk.

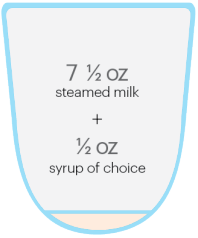
Milk Temperature:
130-140°F



Mocha - *12 oz*
A latte-style hot chocolate with espresso & steamed milk.

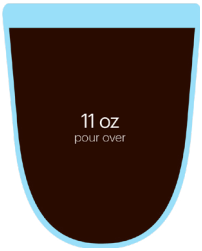
Milk Temperature:
145-155°F

Recommended Product:
Torani Dark Chocolate Sauce



Steamer - *8oz*
Steamed milk, typically with an added syrup flavor (i.e. vanilla). A kid's steamer will typically be in a smaller size, with the milk steamed to only a slightly warm temperature.

Milk Temperature:
110-125°F (Kids) | 145-155°F (Adult)



Pour Over Coffee - *12 oz*
Filtered coffee brewed one cup at a time either manually or by a pour over machine. Usually offered for light roast, single origin coffees and drank black, to highlight the tasting notes of the specific coffee. Uses the same grinder as drip coffee.

Recommended Ratio:
25 g coffee : 400 g (ml) of water

Recommended Pour Time:
2 minutes, 30 seconds

Recommended Drip Time:
1 - 2.5 minutes

Total Brew Time:
3.5 - 5 minutes